Manage users, Groups and Change file permission-30

1. Introduction

In Linux, managing users and groups is important for system administration. File permissions control who can read, write, or execute files. This ensures security and proper resource management in a multi-user environment like RHEL.

2. User Management

Create a User

sudo useradd username

- Creates a new user.
- Example: sudo useradd raj

Set Password

sudo passwd username

- Assigns password for the user.
- Example: sudo passwd raj

Delete a User

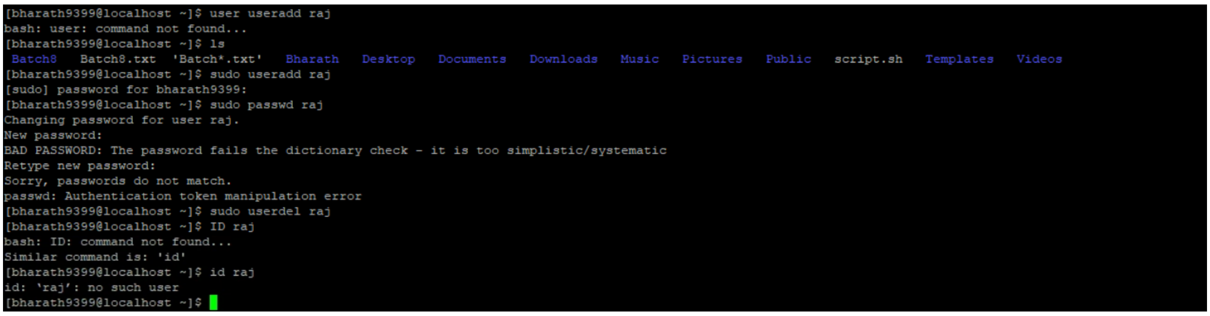
sudo userdel username

- Removes a user from the system.
- Example: sudo userdel raj

View User Information

id username

- Shows UID, GID, and group memberships.


```
[bharath9399@localhost ~]$ user useradd raj
bash: user: command not found...
[bharath9399@localhost ~]$ ls
 Batch8   Batch8.txt  'Batch*.txt'   Bharath   Desktop   Documents   Downloads   Music   Pictures   Public   script.sh   Templates   Videos
[bharath9399@localhost ~]$ sudo useradd raj
[sudo] password for bharath9399:
[bharath9399@localhost ~]$ sudo passwd raj
Changing password for user raj.
New password:
BAD PASSWORD: The password fails the dictionary check - it is too simplistic/systematic
Retype new password:
Sorry, passwords do not match.
passwd: Authentication token manipulation error
[bharath9399@localhost ~]$ sudo userdel raj
[bharath9399@localhost ~]$ ID raj
bash: ID: command not found...
Similar command is: 'id'
[bharath9399@localhost ~]$ id raj
id: 'raj': no such user
[bharath9399@localhost ~]$
```

3. Group Management

Create a Group

sudo groupadd groupname

- Creates a new group.
- Example: sudo groupadd developers

Add User to Group

sudo usermod -aG groupname username

- Adds user to a supplementary group.
- Example: sudo usermod -aG developers raj

Delete Group

sudo groupdel groupname

- Deletes an existing group.

Check Group Membership

groups username

```
[bharath9399@localhost ~]$ sudo groupadd developers
[bharath9399@localhost ~]$ sudo useradd raj
useradd: warning: the home directory /home/raj already exists.
useradd: Not copying any file from skel directory into it.
Creating mailbox file: File exists
[bharath9399@localhost ~]$ sudo usermod -aG developers raj
[bharath9399@localhost ~]$ groups raj
raj : raj developers
[bharath9399@localhost ~]$ sudo groupdel developers
[bharath9399@localhost ~]$
```

4. File Permissions in Linux

Linux uses three types of permissions for files and directories.

- **Read (r)**: Open and read file contents.
- **Write (w)**: Modify file contents.
- **Execute (x):** Run as a program or enter directory.

Ownership

Every file has:

- **User (Owner)**
- **Group**

- **Others**

View Permissions

ls -l

Example output:

-rw-r--r-- 1 raj developers 1234 Aug 20 test.txt

- rw- = Owner (raj) can read and write.
- r-- = Group (developers) can read only.
- r-- = Others can read only.

```
[bharath9399@localhost ~]$ ls -l
total 4
drwxr-xr-x. 2 bharath9399 bharath9399  6 Aug 20 11:49  Batch8
-rw-r--r--. 1 bharath9399 bharath9399 34 Aug 21 10:46  Batch8.txt
-rw-r--r--. 1 bharath9399 bharath9399  0 Aug 21 10:42 'Batch*.txt'
drwxr-xr-x. 3 bharath9399 bharath9399 19 Aug 20 11:51  Bharath
drwxr-xr-x. 2 bharath9399 bharath9399  6 Aug 19 14:22  Desktop
drwxr-xr-x. 2 bharath9399 bharath9399  6 Aug 19 14:22  Documents
drwxr-xr-x. 2 bharath9399 bharath9399  6 Aug 19 14:22  Downloads
drwxr-xr-x. 2 bharath9399 bharath9399  6 Aug 19 14:22  Music
drwxr-xr-x. 2 bharath9399 bharath9399  6 Aug 19 14:22  Pictures
drwxr-xr-x. 2 bharath9399 bharath9399  6 Aug 19 14:22  Public
-rw-r--r--. 1 bharath9399 bharath9399  0 Aug 21 10:50  script.sh
drwxr-xr-x. 2 bharath9399 bharath9399  6 Aug 19 14:22  Templates
drwxr-xr-x. 2 bharath9399 bharath9399  6 Aug 19 14:22  Videos
[bharath9399@localhost ~]$
```

5. Changing Permissions

Using chmod (symbolic method)

chmod u+rwx file.txt # give owner read, write, execute

chmod g+w file.txt # give group write

chmod o-r file.txt # remove read from others

```
[bharath9399@localhost ~]$ touch test.txt
[bharath9399@localhost ~]$ chmod u+rwx test.txt
[bharath9399@localhost ~]$ ls -l 'test.txt'
-rwxr--r--. 1 bharath9399 bharath9399 0 Aug 22 12:03 test.txt
[bharath9399@localhost ~]$ chmod g+w test.txt
[bharath9399@localhost ~]$ ls -l 'test.txt'
-rwxrw-r--. 1 bharath9399 bharath9399 0 Aug 22 12:03 test.txt
[bharath9399@localhost ~]$ chmod o-r 'test.txt'
[bharath9399@localhost ~]$ ls -l 'test.txt'
-rwxrw----. 1 bharath9399 bharath9399 0 Aug 22 12:03 test.txt
[bharath9399@localhost ~]$
```

Using chmod (numeric method)

- Read = 4
- Write = 2
- Execute = 1
- Add them together for full permission.

SUID (Set User ID)

chmod u+s file

File runs with owner's privileges.

SGID (Set Group ID)

chmod g+s directory

New files in directory inherit group ownership.

Sticky Bit

chmod +t /shared

Only file owner can delete files inside shared directory.

umask
Defines default permissions for new files and directories.
Example: umask 022

Examples:

chmod 755 script.sh   # Owner rwx, group r-x, others r-x

chmod 644 notes.txt   # Owner rw, group r, others r

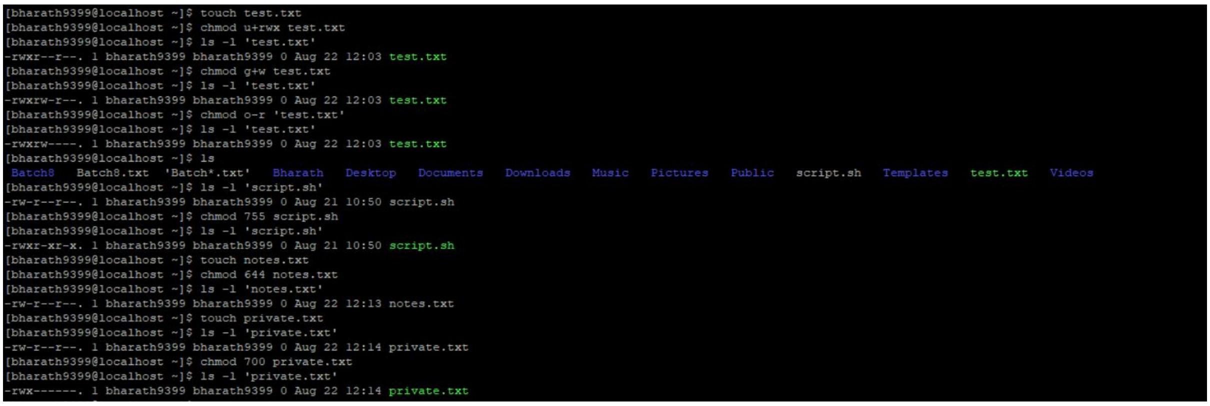
chmod 700 private.txt # Only owner has full access

Changing Ownership

sudo chown user:group file

Example:

sudo chown raj:developers project.doc



```
[bharath9399@localhost ~]$ touch test.txt
[bharath9399@localhost ~]$ chmod u+rwx test.txt
[bharath9399@localhost ~]$ ls -l 'test.txt'
-rwxr--r--. 1 bharath9399 bharath9399 0 Aug 22 12:03 test.txt
[bharath9399@localhost ~]$ chmod g+w test.txt
[bharath9399@localhost ~]$ ls -l 'test.txt'
-rwxrw-r--. 1 bharath9399 bharath9399 0 Aug 22 12:03 test.txt
[bharath9399@localhost ~]$ chmod o-r 'test.txt'
[bharath9399@localhost ~]$ ls -l 'test.txt'
-rwxrw----. 1 bharath9399 bharath9399 0 Aug 22 12:03 test.txt
[bharath9399@localhost ~]$ ls
 Batch8   Batch8.txt  'Batch*.txt'   Bharath   Desktop   Documents   Downloads   Music   Pictures   Public   script.sh   Templates   test.txt   Videos
[bharath9399@localhost ~]$ ls -l 'script.sh'
-rw-r--r--. 1 bharath9399 bharath9399 0 Aug 21 10:50 script.sh
[bharath9399@localhost ~]$ chmod 755 script.sh
[bharath9399@localhost ~]$ ls -l 'script.sh'
-rwxr-xr-x. 1 bharath9399 bharath9399 0 Aug 21 10:50 script.sh
[bharath9399@localhost ~]$ touch notes.txt
[bharath9399@localhost ~]$ chmod 644 notes.txt
[bharath9399@localhost ~]$ ls -l 'notes.txt'
-rw-r--r--. 1 bharath9399 bharath9399 0 Aug 22 12:13 notes.txt
[bharath9399@localhost ~]$ touch private.txt
[bharath9399@localhost ~]$ ls -l 'private.txt'
-rw-r--r--. 1 bharath9399 bharath9399 0 Aug 22 12:14 private.txt
[bharath9399@localhost ~]$ chmod 700 private.txt
[bharath9399@localhost ~]$ ls -l 'private.txt'
-rwx------. 1 bharath9399 bharath9399 0 Aug 22 12:14 private.txt
```

Managing users, groups, and file permissions is key for Linux system security. Correct configuration ensures that only authorized users access files and services. Using useradd, groupadd, chmod, and chown, administrators maintain control over resources.